EUCLID · BOOK I · PROPOSITION 20

Turn two sides into one longer straight length

Extend a side and copy the other onto it. The earlier angle–side comparison then proves the triangle inequality without assuming that a straight route is shortest.

THE PROPOSITION

GREEK Ἑλληνικά

Παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.

LATIN Latine

In quouis triangulo duo latera reliquo maiora sunt quoquo modo coniuncta.

ENGLISH reading translation

In any triangle, any two sides taken together are greater than the remaining side.

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1 Name all three comparisons

GREEK Ἑλληνικά

Ἔστω γὰρ τρίγωνον τὸ ΑΒΓ· λέγω, ὅτι τοῦ ΑΒΓ τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, αἱ μὲν ΒΑ, ΑΓ τῆς ΒΓ, αἱ δὲ ΑΒ, ΒΓ τῆς ΑΓ, αἱ δὲ ΒΓ, ΓΑ τῆς ΑΒ.

LATIN Latine

Sit enim triangulus ΑΒΓ. dico, in triangulo ΑΒΓ duo latera reliquo maiora esse quoquo modo coniuncta, ΒΑ + ΑΓ > ΒΓ, ΑΒ + ΒΓ > ΑΓ, ΒΓ + ΓΑ > ΑΒ.

ENGLISH reading translation

Let ΑΒΓ be a triangle. I say that any two sides together exceed the remaining side: ΒΑ and ΑΓ exceed ΒΓ; ΑΒ and ΒΓ exceed ΑΓ; ΒΓ and ΓΑ exceed ΑΒ.

2 Continue BA beyond A

GREEK Ἑλληνικά

Διήχθω γὰρ ἡ ΒΑ ἐπὶ τὸ Δ σημεῖον,

LATIN Latine

educatur enim BA ad Δ punctum,

ENGLISH reading translation

For let BA be continued to the point Δ,

3 Copy ΓA to AΔ, then join ΔΓ

GREEK Ἑλληνικά

καὶ κείσθω τῇ ΓΑ ἴση ἡ ΑΔ, καὶ ἐπεζεύχθω ἡ ΔΓ.

LATIN Latine

et ponatur AΔ = ΓA, et ducatur ΔΓ.

ENGLISH reading translation

and let AΔ be laid off equal to ΓA, and let ΔΓ be joined.

4 Use the small isosceles triangle

GREEK Ἑλληνικά

Ἐπεὶ οὖν ἴση ἐστὶν ἡ ΔΑ τῇ ΑΓ, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ ΑΔΓ τῇ ὑπὸ ΑΓΔ·

LATIN Latine

iam quoniam ΔA = AΓ, erit etiam ∠ AΔΓ = AΓΔ.

ENGLISH reading translation

Since ΔA equals AΓ, angle AΔΓ also equals AΓΔ.

5 Compare the whole angle with its part

GREEK Ἑλληνικά

μείζων ἄρα ἡ ὑπὸ ΒΓΔ τῆς ὑπὸ ΑΔΓ.

LATIN Latine

itaque ∠ ΒΓΔ > ΑΔΓ.

ENGLISH reading translation

Therefore ΒΓΔ is greater than ΑΔΓ.

6 Use I.19 in the larger triangle

GREEK Ἑλληνικά

καὶ ἐπεὶ τρίγωνόν ἐστι τὸ ΔΓΒ μείζονα ἔχον τὴν ὑπὸ ΒΓΔ γωνίαν τῆς ὑπὸ ΒΔΓ, ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει, ἡ ΔΒ ἄρα τῆς ΒΓ ἐστι μείζων.

LATIN Latine

et quoniam triangulus est ΔΓΒ maiorem habens angulum ΒΓΔ angulo ΒΔΓ, sub maiore autem angulo maius latus subtendit, erit ΔΒ > ΒΓ.

ENGLISH reading translation

Since triangle ΔΓΒ has angle ΒΓΔ greater than ΒΔΓ, and the side opposite the greater angle is greater, ΔΒ is greater than ΒΓ.

7 Replace ΔΑ with its equal ΑΓ

GREEK Ἑλληνικά

ἴση δὲ ἡ ΔΑ τῇ ΑΓ· μείζονες ἄρα αἱ ΒΑ, ΑΓ τῆς ΒΓ·

LATIN Latine

uerum ΔΑ = ΑΓ. itaque ΒΑ + ΑΓ > ΒΓ.

ENGLISH reading translation

But ΔΑ equals ΑΓ. Therefore ΒΑ and ΑΓ together are greater than ΒΓ.

8 Construct the other two comparisons

GREEK Ἑλληνικά

ὁμοίως δὴ δείξομεν, ὅτι καὶ αἱ μὲν ΑΒ, ΒΓ τῆς ΓΑ μείζονές εἰσιν, αἱ δὲ ΒΓ, ΓΑ τῆς ΑΒ.

LATIN Latine

similiter demonstrabimus, esse etiam AB + BΓ > ΓA et BΓ + ΓA > AB.

ENGLISH reading translation

Similarly we shall show that AB and BΓ together exceed ΓA, and that BΓ and ΓA together exceed AB.

9 Conclude the triangle inequality

GREEK Ἑλληνικά

Παντὸς ἄρα τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖξαι.

LATIN Latine

Ergo in quouis triangulo duo latera reliquo maiora sunt quoquo modo coniuncta; quod erat demonstrandum.

ENGLISH reading translation

Therefore, in any triangle, any two sides taken together exceed the remaining side. This is what had to be shown.

Stoicheia · Euclid, Elements I.1–20. Greek and Latin: J. L. Heiberg, Euclidis opera omnia, vol. I (Leipzig, 1883), pp. 2–51; public domain. Latin is Heiberg’s nineteenth-century translation. English reading and notes: Stoicheia, 2026, MIT. Source and edition notes: antonsoo.github.io/stoicheia/